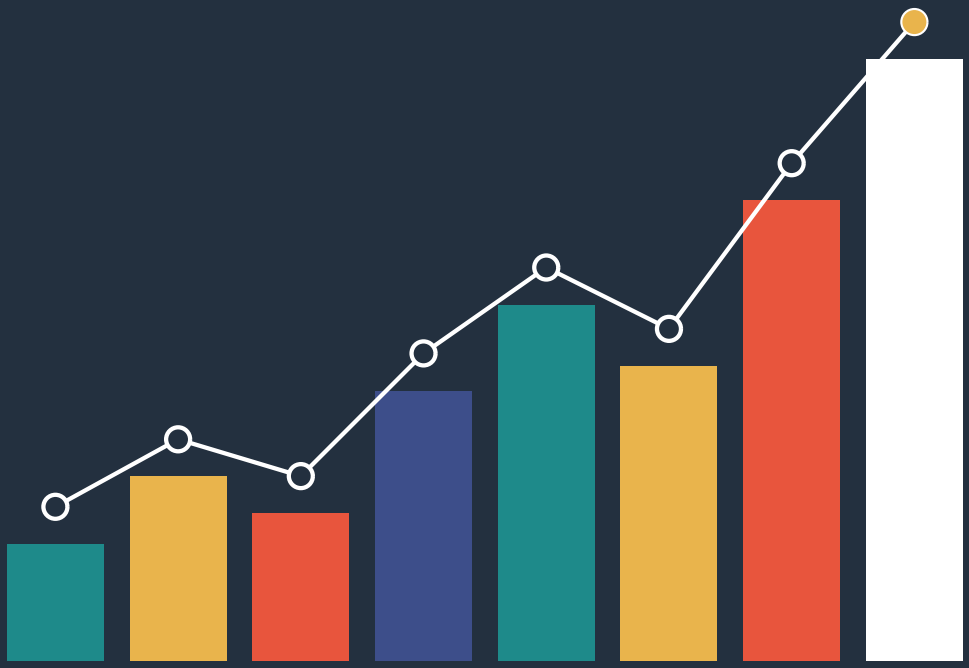
NOTEBOOK No. _____ EDITION 2026



THE ANALYST'S POWERPOINT NOTEBOOK

From your first slide to data-driven storytelling

4 levels | 23 lessons | mini-labs | 1 capstone project

THIS NOTEBOOK BELONGS TO

TRAINER / BATCH

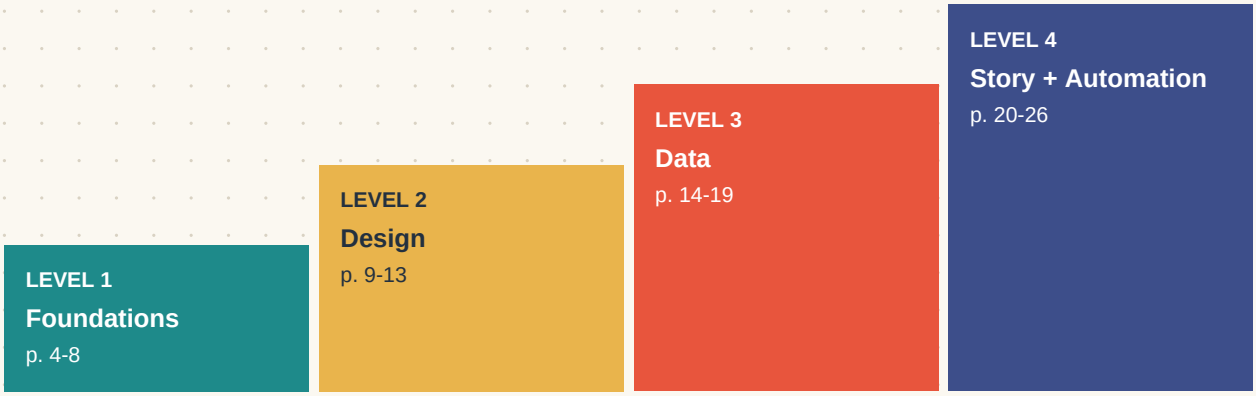
Learn it. Try it. Sketch it. Present it.

WELCOME

Start Here

How this notebook works

This is not a feature manual. It is a **working notebook** built the way data analysts think: start with the message, choose the evidence, then design the slide. Each lesson is one page, so you can finish a lesson in about 30-45 minutes.



Climb one step at a time. Each level assumes only the one before it.

Three habits (look for these boxes)

READ

The page itself. Short rules, then a drawing that shows the rule in action.

MINI-LAB | 3-10 MIN

A hands-on task. Always do it in PowerPoint before you turn the page. Reading is not learning.

ANALYST NOTE

Where a data-analysis habit (clean data, one message, no chartjunk) makes your slide better.

For trainers

- Teach one lesson per session: 10 min explain, 20 min mini-lab, 10 min peer review.
- Swap the sample numbers for your own company or class data so every student gets a different deck.
- Use the capstone (p. 28) as the final assessment and the tracker (p. 29) for sign-off.
- Menu paths are for **Windows / Microsoft 365**. Mac menus are similar; Ctrl usually becomes Cmd.

FIELD NOTE

Fill in the dotted notebook margins with your own shortcuts, colours and questions. The best notebook is a messy one.

MAP

Contents

Four levels, twenty-three lessons

LEVEL 1 | FOUNDATIONS

01 Meet the Workspace 4
02 Slides and Layouts 5
03 Text That Works 6
04 Shapes, Align and Group 7
05 Speed Keys + Practice 1 8

LEVEL 2 | DESIGN

06 Colour Like a Designer 9
07 Type and the Grid 10
08 Pictures and Icons 11
09 Slide Master and Templates 12
10 Motion with Purpose 13

LEVEL 3 | DATA

11 Tables That Read Fast 14
12 Pick the Right Chart 15
13 Build a Chart 16
14 Declutter and Highlight 17
15 The Excel-to-Slide Pipeline 18
16 The KPI Dashboard Slide 19

LEVEL 4 | STORY + AUTOMATION

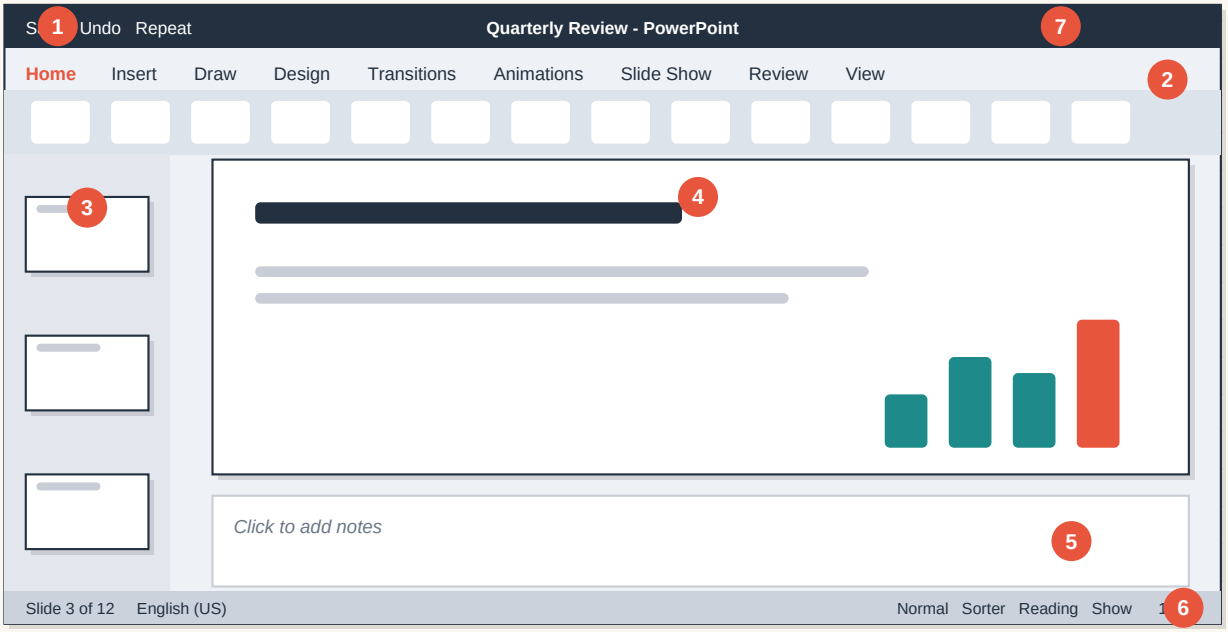
17 Story Structure 20
18 Action Titles 21
19 Navigation and Interactivity 22
20 Power Tools 23
21 Macros with VBA 24
22 Python to Slides 25
23 Present and Accessible 26

\+ Pre-flight checklist (before you present) 27
\+ Capstone project: the 8-slide business review 28
\+ My progress tracker and free notes 29

Meet the Workspace

Know where everything lives before you build anything.

PowerPoint is a stage with a control panel around it. Analysts who know the panel work in half the time. Learn these seven zones once and every lesson after this will make sense.



1 Quick Access Toolbar: save, undo, repeat. Add your favourite tools.

2 Ribbon: commands grouped by job. Tabs change with what you select.

3 Thumbnail pane: your slide sorter. Drag to reorder.

4 Slide canvas: the stage. Anything outside the edge never shows.

5 Notes pane: your speaker script lives here, not on the slide.

6 Status bar: view buttons and zoom. Slide Sorter is great for flow.

7 Search / Tell Me box: type what you want to do (Alt+Q).

MINI-LAB | 3 MIN

Go to **File > Options > Quick Access Toolbar**. Add: **Align Left, Distribute Horizontally, Group, Selection Pane**. These four tools will save you hours this month.

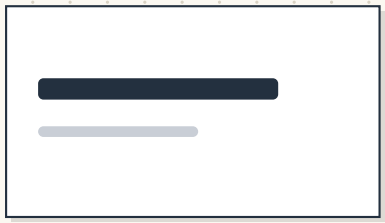
ANALYST NOTE

Treat a slide like a dashboard tile: one message, one visual, one takeaway. The canvas is small on purpose.

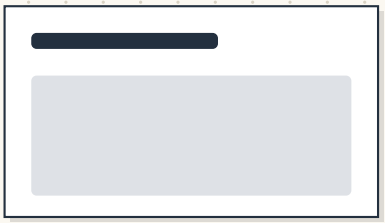
Slides and Layouts

A layout is a promise about where things will sit.

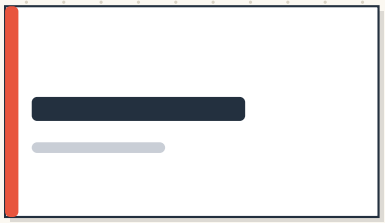
Every slide is built on a **layout**. Layouts contain **placeholders** (the dotted boxes) that hold titles, text, charts and pictures. When you use placeholders, text stays in the same place on every slide, Outline View works, and screen readers can find your titles.



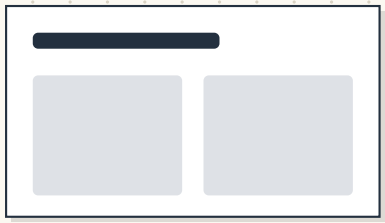
Title Slide



Title and Content



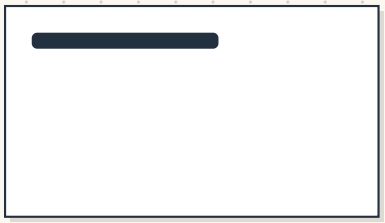
Section Header



Two Content



Comparison



Title Only

How to choose and change a layout

1. **Home > New Slide** (small arrow) lets you pick the layout before the slide is created.
2. **Home > Layout** changes the layout of the slide you are on. Your content moves into the new placeholders.
3. **Home > Reset** snaps every object back to the layout position. Use it after messy dragging.
4. **View > Outline View**: type a title, press Enter for a new slide, press Tab to demote to a bullet.

WATCH OUT

Do not draw a text box for a title. A hand-drawn text box is invisible to Outline View, Accessibility Checker and the slide navigator. Use the title placeholder every time.

MINI-LAB | 5 MIN

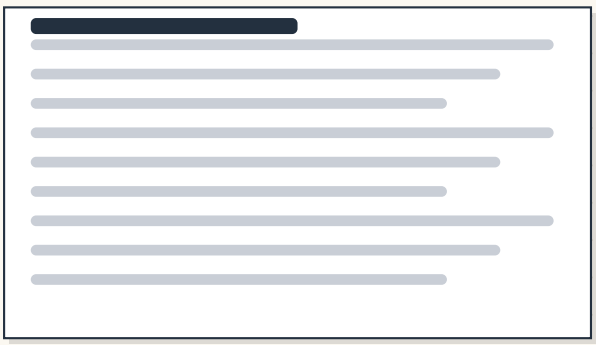
Open **Outline View** and type the titles of a 6-slide story about last quarter's sales. Then use **Layout** to give slide 1 a Title Slide, slide 3 a Section Header and slide 5 a Two Content layout.

Text That Works

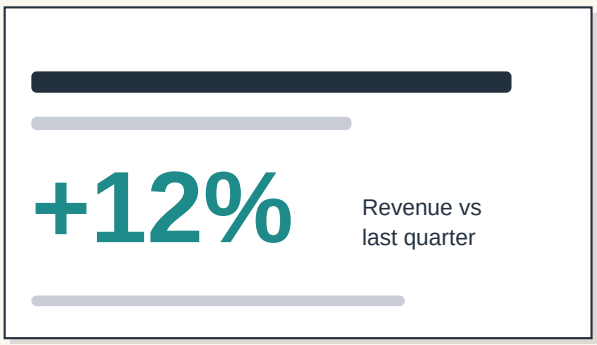
If your audience reads, they are not listening.

Slide text is a **signpost**, not a script. Your voice delivers the detail. Use this size guide for slides that will be projected in a classroom or meeting room.

Element	Size (pt)	Rule of thumb
Slide title	28 - 36	A full sentence that states the finding (see Lesson 18)
Body text	18 - 24	Max 5 lines. Never smaller than 16.
Chart labels	12 - 14	Label directly on the chart where you can
Source / footnote	10 - 11	Always cite the data source



BEFORE: a wall of words



AFTER: one message, one number

Five text rules

- **One slide = one message.** If you need "and also", you need two slides.
- **Three levels maximum:** title, main point, supporting point.
- **Sentence case** is easier to read than ALL CAPS or Title Case Everywhere.
- **Left-align** paragraphs. Centre only short titles or big numbers.
- **The 5-second test:** show your slide to a colleague for five seconds. Can they say what it means?

MINI-LAB | 5 MIN

Find the most text-heavy slide you ever made. Rewrite it as: 1 title sentence + 1 big number or chart + 1 line of context. Move everything else to the **Notes pane**.

Shapes, Align and Group

Alignment is the cheapest way to look professional.

Shapes are your diagram toolkit: boxes, arrows, callouts, icons. The skill is not drawing, it is **lining things up**. Nobody notices perfect alignment, but everybody feels misalignment.



Messy: eyeballed

Clean: Align + Distribute

ANALYST FIELD NOTE

Hold Alt while dragging to turn snapping off for tiny nudges.

The 6-step shape workflow

1. **Insert > Shapes**. Hold **Shift** while dragging for a perfect square or circle.
2. Hold **Ctrl** and drag to duplicate. Add **Shift** too to duplicate in a straight line.
3. Select several objects (Shift+click or drag a box). Open **Shape Format > Align**.
4. Pick **Align Top / Middle / Bottom**, then **Distribute Horizontally** for equal gaps.
5. Check the dropdown: **Align to Selected Objects** (relative to each other) vs **Align to Slide**.
6. Press **Ctrl+G** to group the finished diagram so it moves as one object.

Name what you build

Open the **Selection Pane** (**Alt+F10**) and rename objects: *KPI_Revenue*, *Chart_Trend*. Clear names make animation and accessibility much easier. You can also hide objects with the eye icon.

MINI-LAB | 5 MIN

Draw a 4-step process (Collect, Clean, Analyse, Present) using four rounded rectangles and three arrows. Make it perfectly aligned in under 60 seconds using only the Align tools.

Speed Keys + Practice 1

Hands on the keyboard, eyes on the message.

These are the shortcuts that data analysts use most. Learn two per day. (Windows shown; on Mac, Ctrl is usually Cmd.)

Do this	Press
New presentation	Ctrl + N
New slide	Ctrl + M
Duplicate slide / object	Ctrl + D
Group / ungroup	Ctrl + G / Ctrl+Shift+G
Copy / paste formatting	Ctrl+Shift+C / V
Repeat last action	F4 or Ctrl + Y
Selection Pane	Alt + F10
Hyperlink	Ctrl + K
Search a command	Alt + Q

Do this	Press
Start from slide 1	F5
Start from this slide	Shift + F5
Black / white screen	B / W
Go to slide number	Type number + Enter
Centre / left / right	Ctrl + E / L / R
Font bigger / smaller	Ctrl+Shift+> / <
Paste Special	Ctrl + Alt + V
Nudge a little	Ctrl + Arrow keys
Show guides / grid	Alt+F9 / Shift+F9

PRACTICE 1 | The 10-minute "About Me" deck

Build a 5-slide deck about yourself using **only the keyboard** for new slides, duplicating, grouping and starting the show.

- [ ] Slide 1: title slide with your name and role
- [ ] Slide 2: three facts about you, each with a different icon (Insert > Icons)
- [ ] Slide 3: a 4-shape diagram of your career path, aligned and grouped
- [ ] Slide 4: one big number that matters to you (years of experience, projects, countries)
- [ ] Slide 5: a closing question for the audience
- [ ] Run the show with F5, jump to slide 3, black the screen with B, then Esc

SELF-CHECK

Time yourself. Under 10 minutes means you are ready for Level 2. Over 15 minutes? Redo Lessons 4 and 5 before moving on.

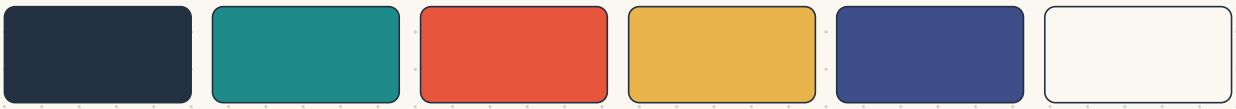
Colour Like a Designer

Grey does the work. One colour does the talking.

Colour is information. Use it to guide the eye, not to decorate. Start with the **60-30-10 rule**: 60% neutral background, 30% secondary tone, 10% accent for the thing that matters.



This notebook's starter palette (copy the hex codes)



Set it once: theme colours

1. **Design > Variants > Colors > Customize Colors**.
2. Paste your hex codes into Accent 1 to Accent 6 (**More Colors > Custom > Hex**).
3. Name the set and save. Now every chart and shape picks up your palette automatically.

Contrast and colour-blindness



- Aim for contrast ratio **4.5 : 1** for body text (check with the free WebAIM contrast checker).
- About 1 in 12 men has red-green colour blindness. Pair **blue + orange**, not red + green, and add labels or shapes.
- For data: grey for context, **one accent** for the point. Max 5-6 colours in a categorical chart.

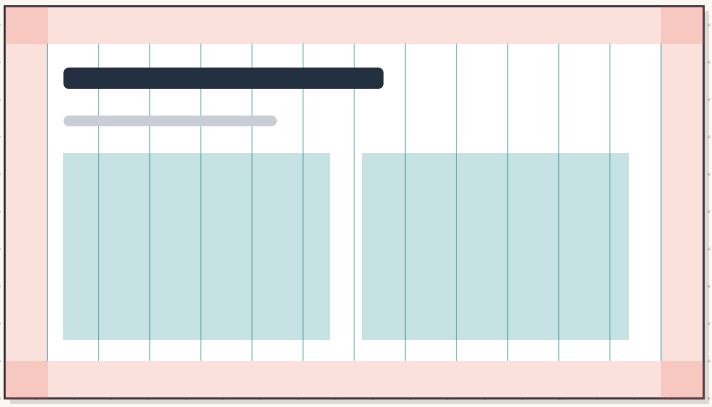
MINI-LAB | 5 MIN

Save the palette as a theme colour set. Then insert a 4-column chart and watch it adopt the new colours.

Type and the Grid

Spacing is a design element. Protect it.

Use **one font family** (or two at most) and let size and weight create hierarchy. Then place everything on a simple grid so your eye always knows where to look next.



12-column grid, 16 px safe margin

- Turn on **View > Guides** and drag them to set margins.
- Turn on **Gridlines** (**Shift+F9**) for a 12-column feel.
- Align text to the same left edge on every slide.
- Leave empty space. White space is not wasted space.

Type scale and pairing

Level	Size	Weight	Use
Title	32-36	Bold	The finding as a sentence
Subtitle	20-24	Regular	Context or scope
Body	18-24	Regular	Max 5 lines
Caption	12-14	Regular, grey	Source, units, notes

- **Safe fonts** that exist on every machine: Calibri, Arial, Segoe UI, Georgia (for headlines).
- Set fonts once: **Design > Variants > Fonts > Customize Fonts**. Never pick fonts slide by slide.
- Line spacing 1.0 to 1.2. Tight enough to read as a block, loose enough to breathe.
- Embed fonts when sharing: **File > Options > Save > Embed fonts in the file**.

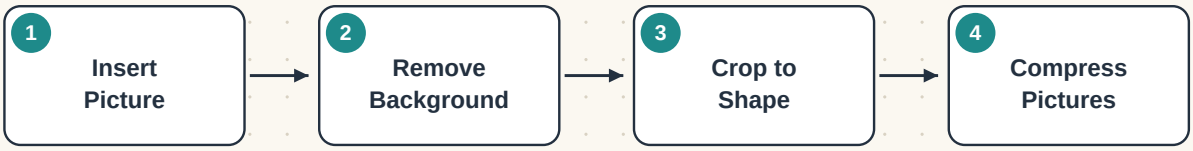
MINI-LAB | 5 MIN

Draw guides at 0.5 inch margins. Place a title, a chart and a takeaway box. Everything must touch a guide or the grid.

Pictures and Icons

One strong image beats five small ones.

Images should **carry meaning**, not fill space. For analysts, that usually means screenshots of tools, icons that label a concept, or one honest photo that makes a number human.



The four-step picture pipeline

- **Insert > Pictures**. Resize with a **corner** handle only. Side handles stretch and distort.
- **Picture Format > Remove Background**, then Mark Areas to Keep / Remove. Good for product shots and logos.
- **Picture Format > Crop > Crop to Shape** gives circular headshots or rounded cards. Use **Fill** or **Fit** to reposition.
- **Compress Pictures** (Picture Format) to 150 ppi for screen. Your 80 MB deck will become 8 MB.

Icons and graphics

- **Insert > Icons** gives you scalable vector icons. Recolour them with Graphics Fill to match your palette.
- Right-click an icon, **Convert to Shape**, to edit individual parts or merge them with other shapes.
- Use **one icon style** (all outline or all solid) across the whole deck.
- Logos on dark backgrounds: use a **transparent PNG** or SVG, never a white-box JPG.

Always add alt text

Right-click a picture, choose **View Alt Text** and write what it shows and why it matters, in one sentence. For decoration only, tick **Mark as decorative**.

MINI-LAB | 7 MIN

Take one photo, remove its background, crop it to a circle and place it beside a big number. Add alt text and compress the file.

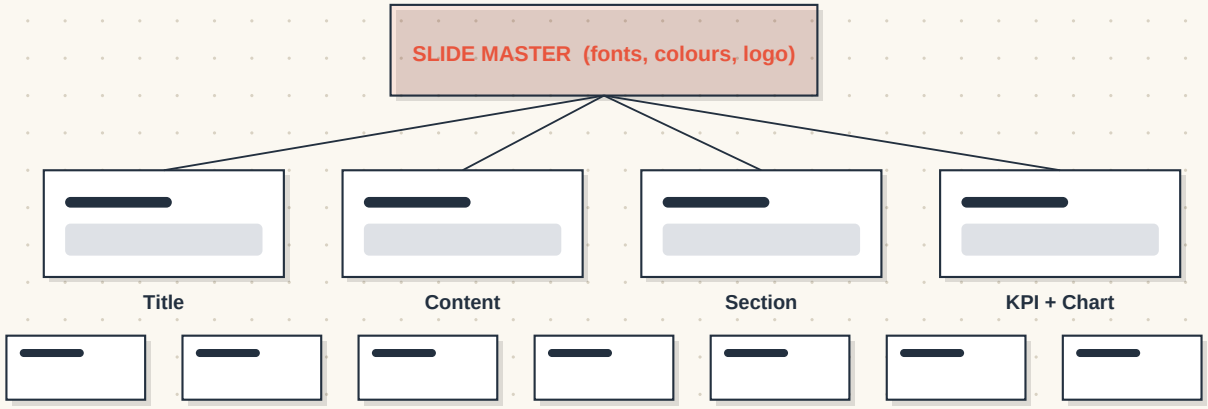
ANALYST FIELD NOTE

Screenshots of your own dashboards are the most credible images a data analyst can use. Crop tight and add a 1-pt border.

Slide Master and Templates

Design once. Reuse forever.

The **Slide Master** is the control room behind every slide. Change a font, colour or logo here and every slide updates. Professional decks are never built slide by slide, they are built on a master.



Your 7 slides inherit everything above

Build your own template

1. **View > Slide Master**. Click the top (largest) thumbnail: this is the master.
2. Set fonts (**Fonts**) and colours (**Colors**) here. Add your logo once in the corner.
3. Select a layout below it and delete the ones you will never use.
4. **Insert Layout** to make a new one, then **Insert Placeholder** (Content, Chart, Picture, Text).
5. Click **Close Master View**. Test using Home > New Slide.
6. **File > Save As > PowerPoint Template (*.potx)** to reuse it forever.

Smart master habits

- Put footers and slide numbers on the master (**Insert > Slide Number**), not on each slide.
- Create a layout named **KPI + Chart**: title on top, 3 small boxes, 1 large chart placeholder.
- Lock your brand: share the .potx and ask your team to start from it, never from a blank file.

MINI-LAB | 10 MIN

Create the **KPI + Chart** layout described above, save it as *MyTemplate.potx* and use it for your capstone project (p. 28).

Motion with Purpose

If an animation does not help understanding, delete it.

Animation is useful in exactly three situations: to **reveal** information step by step, to **show change** between two states, and to **navigate**. Everything else is noise.

Goal	Tool	Where	Analyst example
Reveal in steps	Fade or Appear	Animations > Add Animation	Show one KPI at a time on click
Show change	Morph transition	Transitions > Morph	Bar grows from Q1 to Q2
Navigate	Zoom	Insert > Zoom	Jump to appendix and back
Draw attention	Emphasis (Pulse)	Animations > Emphasis	Pulse the one outlier, once
Slide change	Fade, Push	Transitions	Same transition on every slide

Morph in 5 steps

1. Build slide A: for example three bars at their Q1 heights.
2. Press **Ctrl+D** (in the thumbnail pane) to duplicate the slide.
3. On the copy, change size, position, colour or text of the same objects.
4. Select the second slide, **Transitions > Morph**.
5. Press **Shift+F5**. PowerPoint animates the difference.

Animate a chart by series

Select the chart, add **Wipe** (From Bottom), open **Effect Options** and choose **By Series** or **By Category**. Now each bar group appears on click, matching your spoken story.

Rules of good motion

- Duration **0.3 to 0.6 seconds**. Faster feels cheap, slower feels slow.
- One transition style per deck. No sound effects, no bounce, no spin.
- Use **Animation Pane** (Animations > Animation Pane) to control order and timing.
- Animation does not survive export to PDF. Design every slide so its final state is complete.

MINI-LAB | 8 MIN

Make a Morph between two slides: three circles of the same size on slide 1, then one circle grown large and two shrunk on slide 2. Say aloud what the audience should conclude.

Tables That Read Fast

A table is a lookup tool. A chart is a story tool.

Use a table when the audience must find **exact values**. Use a chart when they must see a **pattern**. If you are unsure, ask: do they need to look up, or to compare?

Sample: a clean slide table

Region	Revenue (M)	Growth YoY	Margin
North	12.4	+8.1%	21.0%
South	9.8	+3.2%	18.4%
East	15.6	+14.7%	24.2%
West	7.1	-2.0%	15.9%
Total	44.9	+6.9%	20.6%

Sample data for practice. Highlight one row; right-align every number.

Make it readable

- **Right-align numbers**, left-align text, keep the same decimals in a column.
- Remove vertical lines. Keep light horizontal lines or gentle banding.
- Maximum about **5 columns x 6 rows** on a slide. More than that belongs in an appendix or an Excel file.
- Highlight the one row or cell that supports your message. Use bold or a soft fill.
- Show units once in the header: *Revenue (M)* rather than repeating "M" in every cell.

Steps in PowerPoint

1. **Insert > Table**, drag out the size you need (or paste from Excel with Keep Source Formatting off).
2. **Table Design**: choose a plain style, untick **Banded Rows** if you do not need them.
3. **Layout > Alignment** sets right-align for numbers; **Cell Margins** adds air.
4. **Table Design > Borders**: delete vertical borders, keep only a thin line under the header.

MINI-LAB | 7 MIN

Reduce a pivot table from a real Excel file to 5 x 4. Then ask: could a chart replace it?

ANALYST FIELD NOTE

Round numbers before showing them. 12.4 beats 12.4172.

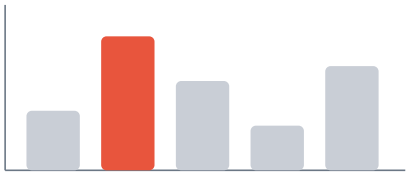
Pick the Right Chart

Start with the question, not with the chart menu.

Every chart answers one question. Write the question first ("Which region is biggest?"), then pick the form that answers it.

Column / bar

Which is bigger?



Compare categories. Sort the bars.

Line

How did it change?



Trends over time. Max 4 lines.

Scatter

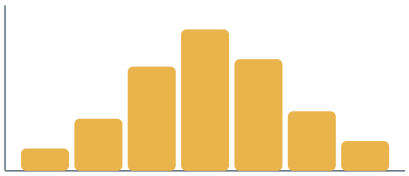
Are X and Y related?



Two numeric variables. Add a trend line.

Histogram

How is it spread?



Distribution: Insert > Chart > Histogram.

Waterfall

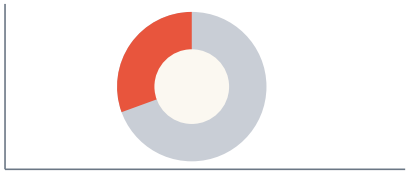
How did we get from A to B?



Bridges: profit, headcount, budget.

Donut (use rarely)

What share is one part?



Only 2-4 parts. Else use bars.

RULES THAT NEVER BREAK

1. Bar charts start at zero. **2.** No 3D. **3.** Avoid dual axes: use two small charts instead. **4.** If values are not parts of a whole, never use a pie. **5.** Label the unit on the axis or the title.

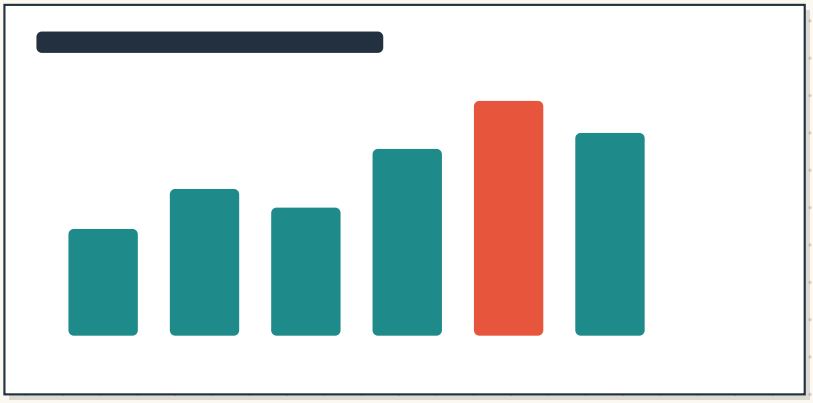
MINI-LAB | 5 MIN

Pick three charts from your own recent work. Write the question each one answers. If you cannot, change the chart.

Build a Chart

Native charts stay editable. Screenshots do not.

Always build charts **inside PowerPoint** (or paste a linked chart from Excel) so you can edit labels, colours and data later. A pasted picture can never be corrected without redoing it.



Build steps

1. **Insert > Chart**, choose type (Column, Line, Scatter...), press OK.
2. A mini spreadsheet opens. Paste your table (headers in row 1, categories in column A). Close it.
3. To edit later: **Chart Design > Edit Data**. Use **Switch Row/Column** if the series are flipped.
4. Click **+ Chart Elements**: remove title (use the slide title), turn on **Data Labels**, remove gridlines.
5. Click a single bar twice, then **Format Data Point** and give the key bar your accent colour.
6. Right-click the chart > **View Alt Text** and write the one-sentence insight.

Sample data to practise with

Month	Jan	Feb	Mar	Apr	May	Jun
Sales (k)	40	55	48	70	88	76
Target (k)	50	50	55	60	70	80

ANALYST NOTE

Clean your data **before** pasting: one header row, no merged cells, numbers stored as numbers. A tidy table gives you a tidy chart.

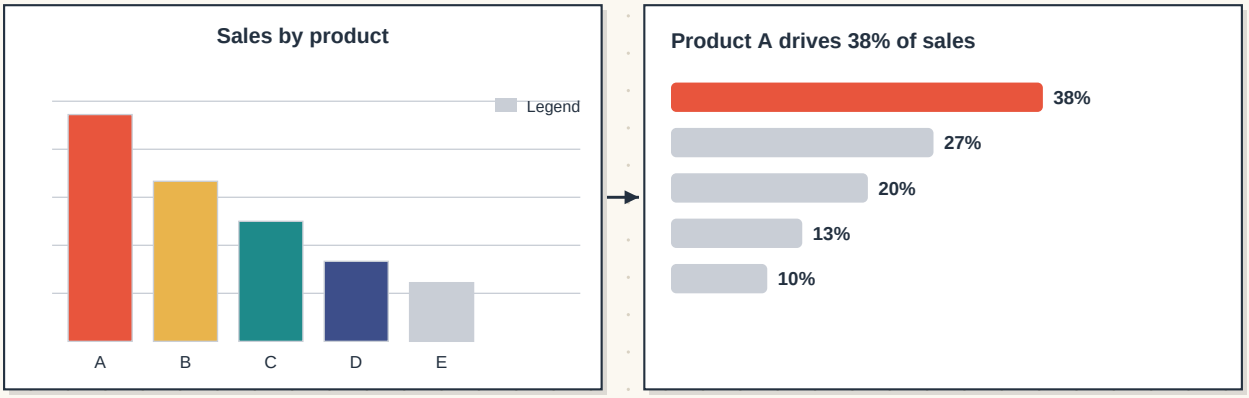
MINI-LAB | 8 MIN

Build a column chart from the sample, highlight the best month in coral, add a Target line using **Change Chart Type > Combo**.

Declutter and Highlight

Remove everything that does not help the message.

Default charts are cluttered. Your job is to **subtract** until only the insight remains. Every gridline, border and colour has to earn its place.



BEFORE: default chart

AFTER: decluttered, one message

The 7 declutter moves

1. **Delete gridlines** and the border around the chart.
2. **Delete the legend**. Label the bars or lines directly.
3. **Turn data labels on** and delete the value axis if labels show the numbers.
4. **Sort** bars from largest to smallest (unless categories have a natural order).
5. **Grey everything**, then colour only the bar that matters.
6. Switch to **horizontal bars** when category names are long.
7. Replace the chart title with an **action title** on the slide (Lesson 18).

Annotate like an analyst

- Add a **text box callout** next to the key point: "Promo launch, 12 May".
- Draw a thin line or arrow, never a thick one.
- Add the **source and date** in 10-pt grey at the bottom-left.

ANALYST NOTE

This is the data-ink idea: if you can remove an element and the message stays the same, it was noise.

MINI-LAB | 10 MIN

Take your ugliest chart from a past report. Apply all seven moves and show before/after to a classmate. Ask: what is the message?

The Excel-to-Slide Pipeline

Pick the paste option on purpose.

As an analyst you live in Excel. Choosing the correct paste option decides whether your slide stays editable, stays current, or breaks when someone moves a file.

Paste option	What happens	Choose when
Use Destination Theme and Embed	Chart copies into the deck with your template colours. Data is stored in the deck.	Default for most charts. Portable.
Keep Source Formatting and Embed	Keeps the Excel look. Data stored in the deck.	You styled it carefully in Excel.
Use Destination Theme and Link Data	Chart follows your theme but stays linked to the workbook.	Monthly reports that refresh.
Picture	A flat image. Not editable.	Final PDF only, or a one-time screenshot.
Keep Text Only	Plain text, no table.	Pasting commentary or numbers into a text box.



Linked data: the monthly routine

1. In Excel, select the chart or range and press **Ctrl+C**.
2. In PowerPoint use **Home > Paste > Paste Special** (Ctrl+Alt+V) and pick a **Link Data** option.
3. Next month: update the Excel file, open the deck, click **Update Links** when asked (or **File > Info > Edit Links to Files**).
4. Before emailing the deck: **Break Link** in the same dialog, or send a PDF, so the recipient cannot see a broken-link warning.

WATCH OUT

Links break when the Excel file is renamed or moved. Keep the deck and workbook in the same shared folder, and never rename the workbook.

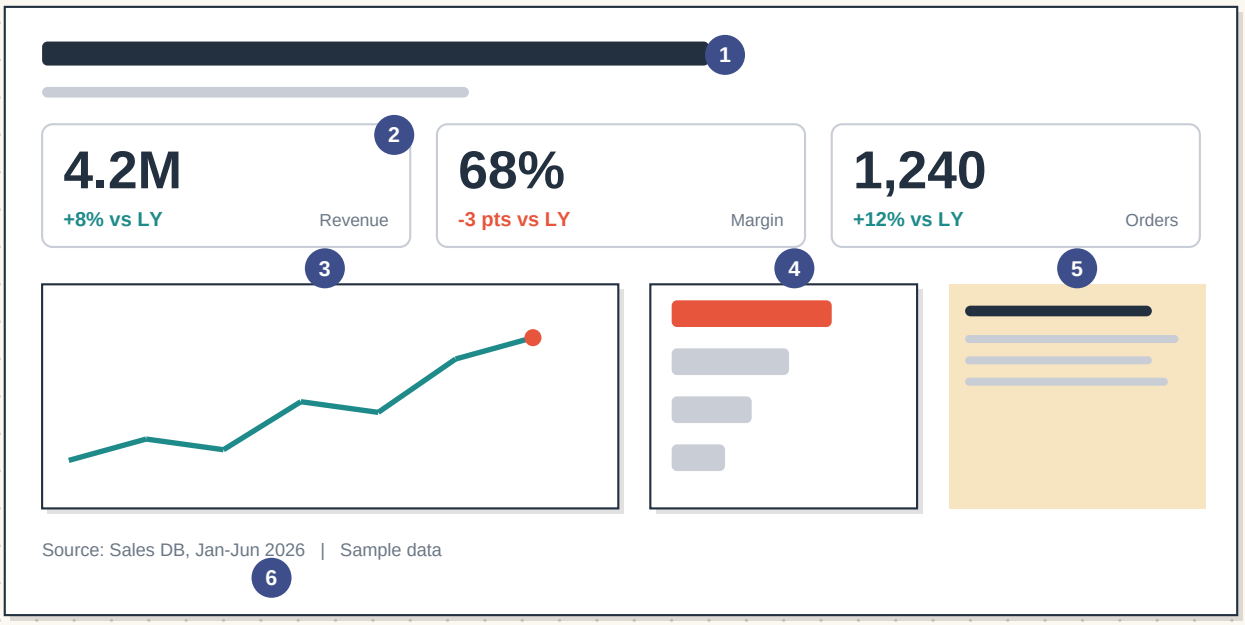
MINI-LAB | 8 MIN

Paste one Excel chart three ways (Embed, Link, Picture). Change a number in Excel and see which versions update.

The KPI Dashboard Slide

The one-slide summary executives actually read.

A dashboard slide shows **3-4 KPIs, 1-2 charts and a takeaway**. Build it once as a layout and reuse it every month.



- **Action title** states the headline ("Revenue grew 8% but margin slipped").
- **KPI cards**: 36-44 pt number, label, and a delta vs a benchmark. Green or teal for good, coral for bad.
- **Trend line** with the latest point highlighted.
- **Ranking bars** with the top item in accent colour.
- **Takeaway panel**: three bullets max, each starting with a verb.
- **Source and date**, always.

MINI-LAB | 12 MIN

Build this slide from the sample data on p. 16 or your own numbers. Duplicate the KPI card with **Ctrl+drag**, then align and group.

Story Structure

Decide what you will say before you open PowerPoint.

The best analysts write the story on paper first. Use **SCQA** to frame it, then the **pyramid principle**: answer first, evidence after.

S SITUATION
What is true and agreed?

"Online sales are 60% of revenue."

C COMPLICATION
What changed or is wrong?

"Conversion fell from 3.1% to 2.4%."

Q QUESTION
What must we decide?

"Where are we losing customers?"

A ANSWER
What do you recommend?

"Fix checkout on mobile; expect +0.5 pt."

Storyboard on paper (sketch here)

SLIDE 1	SLIDE 2	SLIDE 3
message: _____	*message:* _____	*message:* _____

SLIDE 4	SLIDE 5	SLIDE 6
message: _____	*message:* _____	*message:* _____

A reliable 8-slide shape

- 1 Title | 2 Executive summary (the answer) | 3 Context | 4-6 Evidence (one finding each)
- 7 Recommendation and impact | 8 Next steps and owners

MINI-LAB | 10 MIN

On paper, with no computer, write the SCQA for your capstone topic, then sketch six frames. Share it with a partner for feedback before you build.

Action Titles

Titles are the story. The slide is the proof.

A **topic title** labels the slide. An **action title** states the conclusion. Read only your titles in Outline View: they should tell the whole story on their own.

Weak (topic)	Strong (action title)
Sales by region	East grew 14.7% and now leads all regions
Customer churn	Churn doubled after the March price change
Marketing spend	Paid search delivers 3x the return of social
Q2 results	Q2 beat target by 6%, driven by repeat customers
Employee survey	Satisfaction is high, but only 41% see a career path
Recommendation	Pause the West store rollout until margin recovers above 18%

A formula that always works

Subject + verb + number + so-what. *"Mobile conversion + fell + 0.7 points + because checkout is slow."*

- Keep titles to **two lines maximum** at 28-32 pt.
- Put the number in the title; the chart then proves it.
- Use plain verbs: grew, fell, doubled, lags, beats, drives.
- Do not start with "Overview of", "Summary of" or "Analysis of".

The title-only test

1. Open **View > Outline View**.
2. Read only the titles aloud, top to bottom.
3. Does a colleague understand the full recommendation? If not, rewrite titles, not slides.

MINI-LAB | 8 MIN

Rewrite five topic titles from an old deck into action titles with numbers. Then check them in Outline View.

ANALYST FIELD NOTE

If you cannot write the action title, you do not yet know the insight. Go back to the data.

Navigation and Interactivity

Let the audience steer, without getting lost.

Not every talk is linear. Questions come, and you may need to jump to the appendix and back. Build navigation that makes you look prepared, not lost.

Feature	Where	Use it for
Sections	Home > Section > Add Section	Group slides (Context, Evidence, Next steps); collapse in the thumbnail pane
Summary / Section Zoom	Insert > Zoom	A visual menu slide; click a tile, return automatically
Hyperlink to slide	Ctrl + K > Place in This Document	Agenda items, appendix links
Action button	Insert > Shapes > Action Buttons	Home, Back and Next buttons on every slide
Trigger	Animations > Trigger > On Click of	Click an object to reveal a hidden answer
Hide slide	Slide Show > Hide Slide	Backup slides that appear only on request

Build a clickable agenda in 6 steps

1. Make four tiles (rounded rectangles) on one slide: Context, Evidence, Recommendation, Appendix.
2. Add each section with **Home > Section > Add Section** and name it.
3. Select the tile slide, **Insert > Zoom > Section Zoom** and choose all four sections.
4. In **Zoom Format** tick **Return to Zoom** so you land back on the menu after each section.
5. Draw a small "home" action button on the master so every slide has a way back.
6. Test with F5. Click every tile. Hide the Appendix tile if you want it as a secret backup.

ANALYST NOTE

Keep an "Appendix" section with methodology, definitions and extra cuts of the data. Having it hidden but ready is what makes a presenter credible.

MINI-LAB | 10 MIN

Turn your capstone outline into a Section Zoom menu. Then run it and practise jumping to the appendix in two seconds.

Power Tools

Ten features most people never open.

These tools turn a decent deck into a polished one. Learn them one by one; each takes under a minute.

Tool	Where	What it does for you
Merge Shapes	Shape Format > Merge Shapes	Union, Combine, Fragment, Intersect, Subtract: draw custom icons and cut-outs
Eyedropper	Shape Fill > Eyedropper	Pick an exact colour from a logo or any pixel on screen
Format Painter	Home (brush icon)	Copy style to many objects; double-click to keep it active
Edit Points	Right-click shape > Edit Points	Reshape lines and freeforms to trace any outline
Selection Pane	Alt + F10	Rename, reorder and hide objects; fix accessibility reading order
Guides and Grid	View > Guides, Gridlines	Precise alignment with snapping to the grid
Slide Size	Design > Slide Size	Switch 16:9 to 4:3 or A4 for print; use Custom for posters
Insert Screenshot	Insert > Screenshot > Screen Clipping	Capture a dashboard region without leaving PowerPoint
Designer (Microsoft 365)	Home > Designer	Layout suggestions for pictures and text; use as ideas only
Reuse Slides	Home > New Slide > Reuse Slides	Pull slides from another deck, with or without its design

Merge Shapes in action



Union Subtract Intersect

ANALYST FIELD NOTE

Hold Shift when you click Format Painter or double-click it for multiple targets.

MINI-LAB | 8 MIN

Use Merge Shapes to subtract a small circle from a large one and make a donut. Then colour it with the Eyedropper from your logo.

Macros with VBA

Teach PowerPoint to do your repetitive work.

A **macro** is a small program stored in your deck. Use macros for boring tasks: renaming, resizing, exporting. Macros need a **.pptm** file (macro-enabled presentation), and you should only run macros you wrote or trust.

Your first macro

1. Press **Alt + F11** to open the VBA editor (enable the Developer tab in Options if needed).
2. **Insert > Module**, paste the code below, then press **F5** with the cursor inside a Sub to run it.
3. Save as **PowerPoint Macro-Enabled Presentation (.pptm)**.

Macro 1: print every slide title to the Immediate window (Ctrl+G).

```
Sub ListSlideTitles()
    Dim s As Slide
    For Each s In ActivePresentation.Slides
        If s.Shapes.HasTitle Then
            Debug.Print s.SlideIndex & ": " & s.Shapes.Title.TextFrame.TextRange.Text
        End If
    Next s
End Sub
```

Macro 2: apply one font to every text box in the deck.

```
Sub SetAllFonts()
    Dim sld As Slide, shp As Shape
    For Each sld In ActivePresentation.Slides
        For Each shp In sld.Shapes
            If shp.HasTextFrame Then shp.TextFrame.TextRange.Font.Name = "Calibri"
        Next shp
    Next sld
End Sub
```

Macro 3: export all slides as PNG images into a "slides" folder next to the file.

```
Sub ExportPNGs()
    Dim p As String
    p = ActivePresentation.Path & "\slides"
    If Dir(p, vbDirectory) = "" Then MkDir p
    ActivePresentation.Export p, "PNG"
End Sub
```

WATCH OUT

Macros are blocked in files downloaded from the internet and are not available in PowerPoint for the web. Always test on a copy of your deck.

MINI-LAB | 10 MIN

Run Macro 1 on your capstone deck. Read the titles in the Immediate window: do they tell the story (Lesson 18)?

Python to Slides

Your analysis, delivered straight into a deck.

If you already use Python, **python-pptx** can build slides directly from a DataFrame. That means a monthly report can be rebuilt in seconds instead of rebuilt by hand. It is the data analyst's superpower.



Install and build

```
# pip install python-pptx
from pptx import Presentation
from pptx.util import Inches
from pptx.chart.data import CategoryChartData
from pptx.enum.chart import XL_CHART_TYPE

prs = Presentation()
slide = prs.slides.add_slide(prs.slide_layouts[5])   # Title Only layout
slide.shapes.title.text = "Sales rose 12% in Q3"

cd = CategoryChartData()
cd.categories = ["Q1", "Q2", "Q3"]
cd.add_series("Sales (M)", (10, 12, 14))
slide.shapes.add_chart(XL_CHART_TYPE.COLUMN_CLUSTERED,
                       Inches(1), Inches(1.5), Inches(8), Inches(5), cd)
prs.save("report.pptx")
```

Ideas for your own pipeline

- Start from your company **.potx template**: `Presentation("MyTemplate.potx")` keeps your brand.
- Loop over regions and create **one slide per region** with its own chart.
- Pull the number from pandas into the title: `f"Sales rose {pct:.0f}% in Q3"`.
- Add the source line and date automatically so no report goes out without them.

ANALYST NOTE

Automate only after the manual design works. A template you have refined in Lessons 6-9 will make the automated deck look professional.

MINI-LAB | 15 MIN

Run the script above, open report.pptx, then change the data and re-run. Next, loop over three products and make three slides.

Present and Accessible

A great deck still needs a great delivery.

Your slides support you. They do not replace you. Prepare the room, the script and the access needs of every viewer.

Deliver

- Write a 1-2 sentence script per slide in the **Notes pane**.
- **Slide Show > Rehearse Timings** to measure your talk.
- Use **Presenter View** to see notes and next slide while the audience sees the slide.
- Press **B** for a black screen when you want eyes on you.
- Open the deck at least 15 minutes before; test the projector and fonts.
- Pause 3 seconds after each key number.

Include everyone

- Run **Review > Check Accessibility** and fix every error.
- Add **alt text** to charts and images; mark decoration as decorative.
- Reading order = **bottom-to-top** in the Selection Pane. Arrange it logically.
- Do not rely on colour alone; add labels or patterns.
- Use real **table headers** and title placeholders on every slide.
- Minimum 18 pt text, strong contrast.

Share and export

Need	Do this
Email to a client	**File > Export > Create PDF/XPS**. Check Options > Optimise for standard.
Handout for class	**File > Export > Create Handouts**, or PDF with 3 slides per page and notes lines.
Run unattended	**Slide Show > Set Up Slide Show > Browsed at a kiosk**, or Export as video (MP4).
Team edits	Save to OneDrive or SharePoint, then **Share** and co-author live.
Security	**File > Info > Inspect Presentation** removes hidden notes, comments and metadata.

MINI-LAB | 10 MIN

Run the Accessibility Checker on your capstone, fix everything it finds, and practise the opening 30 seconds aloud.

QUALITY GATE

Pre-flight Checklist

Tick every box before you press send or present.

Pilots use checklists because it is easy to forget small things. So do analysts. Keep this page next to your keyboard and copy it for every deck.

STORY

- [ ] One clear message per slide
- [ ] Titles are action titles with numbers
- [ ] Executive summary comes first
- [ ] Every chart answers one question
- [ ] Next steps and owners are stated

DESIGN

- [ ] Fonts and colours come from the theme
- [ ] Max 2 fonts, 18 pt minimum body
- [ ] Everything aligned to the grid
- [ ] Consistent margins and spacing
- [ ] No stretched or pixelated images

DATA

- [ ] Numbers match the source file
- [ ] Bars start at zero, no 3D
- [ ] Units, date range and source shown
- [ ] Gridlines and legends removed
- [ ] Key point highlighted in one accent

ACCESS + FILE

- [ ] Alt text on charts and images
- [ ] Accessibility Checker shows no errors
- [ ] Links tested or broken on purpose
- [ ] Hidden notes and comments cleaned
- [ ] Spell-check run; PDF copy saved

Final five-minute routine

1. Read the titles only (Outline View).
2. Press F5 and click through once without speaking.
3. Compare 3 numbers to the source.
4. Run Check Accessibility.
5. Save as .pptx and as PDF.

FINAL PROJECT

Capstone Project

The 8-slide business review.

Brief: you are the analyst for a retail company. Build an 8-slide review for management that explains why **profit fell in Q2** and recommends what to do. Use the sample data below, or replace it with your own for an original project.

Month	Revenue (k)	Cost (k)	Profit (k)	Orders	Returns %
Jan	410	330	80	1,180	4.1
Feb	395	318	77	1,150	4.3
Mar	440	350	90	1,260	4.0
Apr	455	395	60	1,310	5.2
May	470	420	50	1,350	6.0
Jun	462	428	34	1,330	6.8

Sample data: costs and returns rose faster than revenue.

Required slides

#	Slide	Must include
1	Title	Your name, date, template from Lesson 9
2	Executive summary	Answer first, 3 bullets, one number
3	Context	KPI dashboard slide (Lesson 16)
4-6	Evidence	Three decluttered charts, each with an action title; one table
7	Recommendation	Two options, impact estimate, risk
8	Next steps	Owners, dates, appendix menu with Zoom

Marking rubric (100 points)

Area	Points	Look for
Story and action titles	25	Titles alone tell the story
Data accuracy and chart choice	25	Right chart, correct numbers, sources cited
Design (theme, grid, colour)	20	Consistent, aligned, restrained
Accessibility and file quality	15	Alt text, checker passes, PDF saved
Delivery (3-minute talk)	15	Clear opening, timing, notes

Stretch goals: generate one slide with python-pptx (Lesson 22); add a macro that lists the titles (Lesson 21).

LOG BOOK

My Progress and Notes

Colour in a box each time you finish a lesson.

LEVEL 1 | FOUNDATIONS — Lessons 1-5

1	2	3	4	5

LEVEL 2 | DESIGN — Lessons 6-10

6	7	8	9	10

LEVEL 3 | DATA — Lessons 11-16

11	12	13	14	15	16

LEVEL 4 | STORY + AUTOMATION — Lessons 17-23

17	18	19	20	21	22	23

Capstone submitted: _____ / _____ / 2026

Trainer sign-off: _____

My shortcuts, colours and questions

LAST NOTE

Go build something that makes a number impossible to ignore.